

Mapping Land Cover Change in Odzala-Kokoua National Park

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Research Question:

How has the park been impacted by mining and agriculture?

Background:

Odzala-Kokoua National Park (designated in 1935) lies in the heart of the Republic of Congo, covering approximately 13,546 sq. km. The park's African rainforest accounts for 18% of the world's remaining rainforest and is recognized as one of the world's largest carbon sinks. Additionally, the park contains a variety of land cover types that support thousands of species of birds, mammals, fish, plants, and other living creatures.

Over time the park has become a hotspot for mining, agriculture, and other anthropogenic activities resulting in negative impacts to the park. This project aims to assess the extent and spatial distribution of mining and agricultural activity within the park.

Data and Imagery:

Datasets:

Our project sponsor and point of contact Gwili provided multiple datasets to assist us with our classification.

- GFW_Mines
- Mines_SemiIndustrial_2021to202405
- Minesartisanales202
- OdzalaSurveyData_2021to2023_GG_20260202
- Odzala_AOIs_20260202
- OKNP_AOI
- OKNP_202409

To further assist with identifying anthropogenic activity we utilized the Meta Buildings dataset that allowed for quick identification of human activity.

Imagery:

The basis of our project was Sentinel-2 imagery, we utilized four separate L2A Sentinel-2 scenes to cover our area of interest. Four scenes were selected based on the least amount of cloud cover to reduce atmospheric interference. Each scene covers roughly a quarter of our area of interest, three of the four scenes are dated April 4, 2025 while the North West scene is dated December 12, 2025. This temporal difference presented challenges in classifying the land in a uniform manner due to seasonal and phenological differences.

S2_10m_OdzalaNW_12Dec2025

S2_10m_OdzalaNE_4Apr2025
S2_10m_OdzalaSW_4Apr2025
S2_10m_OdzalaSE_4Apr2025

To supplement our Sentinel-2 scenes we used three Planet basemaps with a finer spatial resolution which allowed for deeper analysis and confirming land cover classification in areas where the Sentinel-2 scenes exhibited excessive cloud cover. The planet basemaps we used are from February 2025

In addition to these images, after our classification was complete and we successfully identified several new areas of interest we used planet scenes from different time periods to demonstrate the change in mining area over time. For our main mining area of interest we found four scenes (November 14, 2016, March 6, 2017, April 20, 2021, and May 6, 2025). Identifying quality scenes to use for this temporal analysis was quite difficult as most of the older scenes had high cloud cover and low overall quality.

Finding high quality imagery in this region proved to be a significant challenge as the Democratic Republic of Congo lies close to the equator in the Intertropical Convergence Zone, characterized by frequent atmospheric disruption and movement. The rainforest also releases large amounts of moisture, contributing to persistent cloud cover.

Another imagery challenge was addressing the challenge of classifying an area that is actively being developed. The prominent mining and anthropogenic activity in the area necessitated imagery from various times to fully address the extent of mining and agriculture in Odzala Kokoua national park.

Methodology & Classification Process:

Defining Classes:

For our area of interest, the main land cover types were forest, savanna, bright bare earth, dark bare earth, water, clouds, and shadows. By using higher level classes for our classification, we could mask out the forest representing the majority of our area of interest and then isolate land cover types common to mining and agriculture such as bright and dark bare earth.

Training Data:

Manually collecting training data was one of the most time-intensive aspects of the project. To power our classifications, we created polygons representative of the classes that would be most representative of the land cover types within the park (forest, bright_barren, dark_barren, clouds, shadows, water, and savanna). Each polygon was drawn manually in ArcGIS Pro or in eCognition for our later classification and the land cover types were

confirmed using our Sentinel-2 imagery, Planet basemaps, and the survey data provided to us by our sponsor.

Separability Scores:

Our separability scores showed that the most difficult classes to distinguish between were clouds and barren_bright. Additionally, mapping barren_dark and savanna areas presented lower Jeffries-Matusita separability scores.

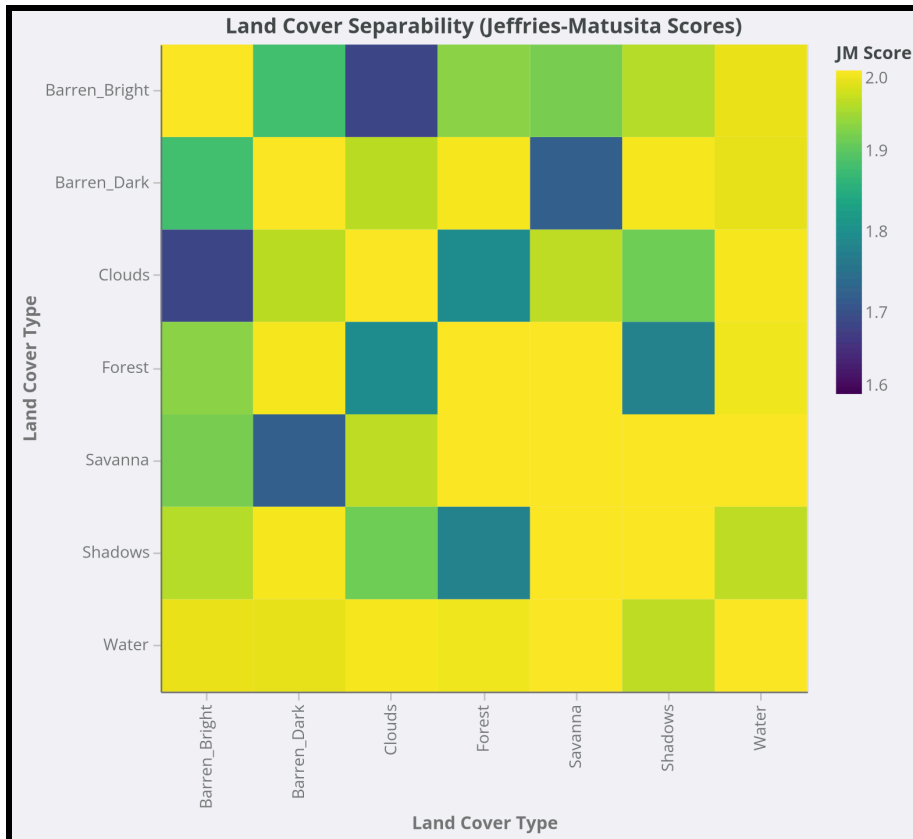


Chart identifying the separability scores for our land cover types using color to represent Jeffries-Matusita scores.

Pixel-Based Classification:

After collecting training data we ran our first pixel-based classification in ENVI. The first classification was a supervised classification using the training data that we had collected and the maximum likelihood algorithm. Post-processing for the classification included smoothing the image and aggregating smaller features to reduce noise. This classification mapped our main land cover types (forest, bright_barren, dark_barren, clouds, shadows, water, and savanna) however, failed to sufficiently map all types across the four tiles as there was an uneven distribution of training data across the four Sentinel tiles which led to high variance in the classification between the tiles.

To correct this error, we collected more training data, ensuring to collect enough features from each tile. After supplementing our training dataset we ran the supervised classification

again on each tile individually and received a much better result. Once again smoothing and aggregation of smaller features was applied during post-processing to remove noise and provide a clearer classification map.

Object-Based Classification:

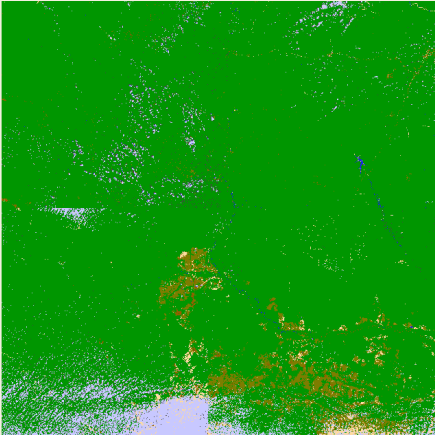
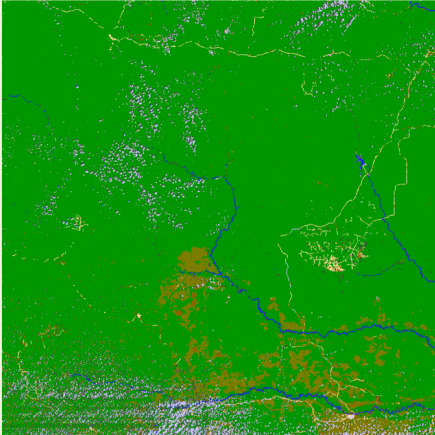
An object-based classification was conducted using eCognition. This classification involved segmenting our imagery using the multiresolution segmentation algorithm with a scale = 100, shape = 0.5, and compactness = 0.5. After segmenting the image, samples were collected for each of our classes (forest, bright_barren, dark_barren, clouds, shadows, water, and savanna) and the classification was run. This process was repeated for each of our four Sentinel-2 images to ensure accuracy and account for differences between the images. This classification was much more effective in identifying our major classes and had significantly less noise than the pixel-based classification.

Temporal Analysis:

After classifying our imagery we were able to isolate specific areas of interest for mining and agricultural activity. We created polygons for these important areas and used them to collect multiple high resolution planet images from different points in time. These images would be used to highlight the impact that mining has on the land by showing the increase in mining area and decrease in forest area over time. After collecting the imagery, we classified each one using a supervised classification in ENVI with the maximum likelihood algorithm. After classifying all of the images, we could then compute statistics and determine the change in land cover for each year.

Results & Accuracy Assessment:

Classification Comparison:

Pixel-Based Classification	Object-Based Classification
	

Classification Area Comparison Table

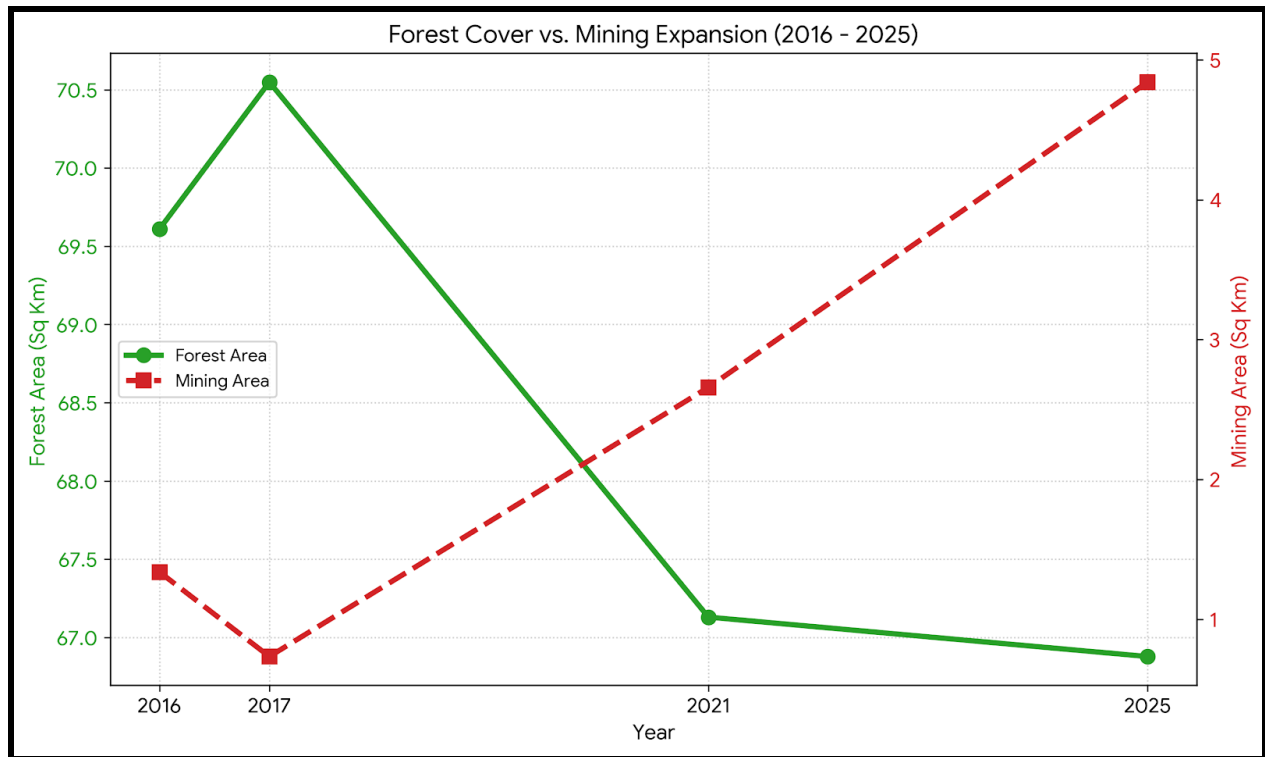
Class Name	Object-Based Area (Sq Km)	Pixel-Based Area (Sq Km)	Difference (Sq Km)	% Change
Forest	41638.4	39283.6	+2,354.80	+5.99%
Savanna	2059.42	1378.7	+680.72	+49.37%
Clouds	1182.91	2181.59	-998.68	-45.78%
Shadows	712.51	450.89	+261.62	+58.02%
Barren_Dark	239.13	174.45	+64.68	+37.08%
Barren_Bright	183.35	495.62	-312.27	-63.01%
Water	71.89	55.37	+16.52	+29.84%

Accuracy & Validation:

Visually, the object-based classification map appears accurate and successfully identifies most of the park's major features. The pixel-based classification appears less accurate as it struggled to see through cloud coverage and outline major park features such as water and small mining areas. To validate results and assess accuracy of the object-based classification 30 random points were generated and individually checked for accuracy. Each point's predicted class was tracked and then verified using Google Earth high resolution imagery. Out of 30 points, 29 were classified correctly, giving an accuracy percentage of 96.66%. After comparing our classification with the Google Earth imagery we concluded that the cloud, shadow, and bright bare earth classes are the most difficult to classify. While the overall classification appears visually correct, zooming in to mixed land coverage areas can occasionally reveal small areas with incorrect classification.

Mining Impact:

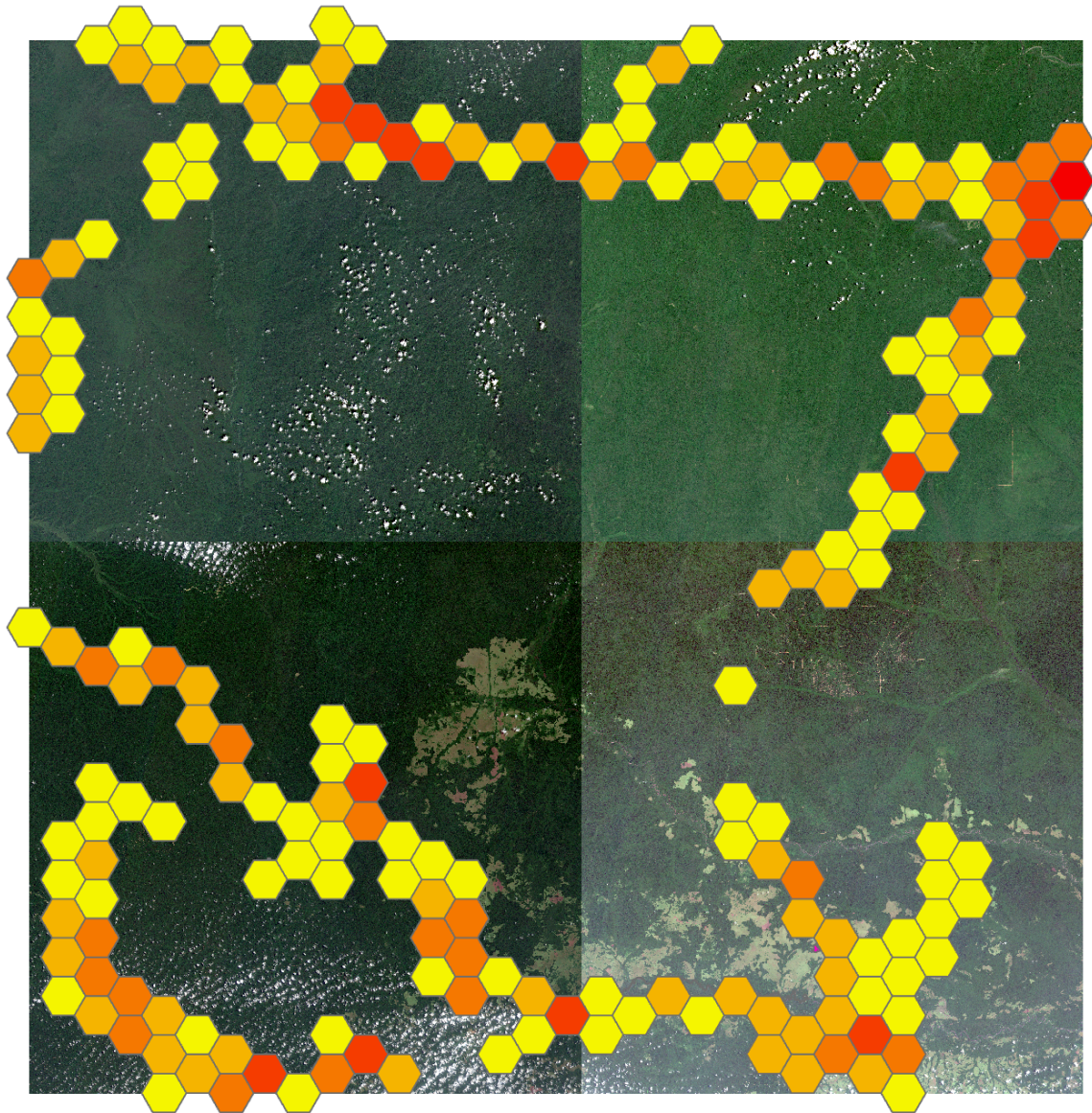
The time series analysis on the main mining AOI was conducted using high resolution Planet imagery from 2016, 2017, 2021, and 2025. Each image was individually classified to calculate mining and forest cover areas. After classifying the change in area over time was calculated for each image. Our main mining area of interest studied between 2016 and 2025 showed a total mining expansion of 3.50 Sq Km (+261.6% Growth) and a total forest loss of -2.73 Sq Km (-3.9% Loss).



Graph showing the change of forest and mining areas over the years 2016, 2017, 2021, and 2025.

Agriculture Hotspots:

Agriculture is a secondary concern for the park. Our analysis found that agricultural activity is most concentrated near villages and follows the road network. To identify hotspots, we filtered all dark barren earth classifications to show only small patches within a three-kilometer buffer of the road system, then created a tessellation over the scenes and calculated the dark barren area within each hexagon. The results confirm that slash-and-burn agriculture is most active at the park's margins and in areas with road access.



Map of agriculture hotspots surrounding Odzala-Kokoua National Park.

Challenges:

Throughout this project we ran into a number of difficulties associated with classifying the land in Odzala-Kokoua national park. The most difficult challenge with this project was finding high resolution cloud-free imagery. The park's geographic location within the Intertropical Convergence Zone — an area characterized by intense atmospheric activity — made finding cloud-free imagery extremely difficult. Another challenge was specifically isolating both mining and agriculture areas. The most effective method we found for identifying mining areas was isolating the bright bare earth areas and closely examining them for mining patterns.

Slash-and-burn agriculture common to the region is characterized by three unique stages each creating a different spectral signature; these varying stages made identifying and classifying agricultural land particularly difficult. To address these challenges, we focused on

identifying hotspots of agriculture instead of identifying the exact location of each field. Recently cleared fields were usually classified as dark barren and could easily be identified as agriculture because they tended to be much smaller than non-agriculture-related burn scars and were usually within about three kilometers of a road. To create a hotspot map, we first filtered all the dark barren earth to show only small patches of dark barren earth within a three-kilometer buffer of the road system. We then created a tessellation over the scenes and calculated how much area classified as dark barren was in each hexagon.

The massive area of interest was also an issue as data management and processing times proved to be quite a challenge with software frequently crashing and long processing times.

Future Improvements:

Looking forward, there are three improvements we would prioritize. The first improvement would be running another object-based classification in eCognition using different parameters. Our primary classification produced strong results, by tweaking the compactness and shape parameters, an even better classification may be possible. Another improvement for this map would be collaboration with a team located in the park to collect training data and create more accurate polygons. By using in-situ data the classification could improve detail and accuracy measures. The final improvement would be reclassifying the data with four images from the same date. The struggle to find cloud free imagery resulted in three of our main Sentinel-2 tiles being from April and the last tile being from December. There are small differences between the two and seasonal changes may cause misclassification or inconsistencies.

Conclusion:

This project successfully mapped land cover change in Odzala-Kokoua National Park using Sentinel-2 satellite imagery, high-resolution Planet imagery, and a combination of both pixel-based and object-based classification methods. The object-based classification produced a highly accurate result (96.66%) and clearly identified the spatial distribution of forest, savanna, bare earth, water, and other land cover types across the park.

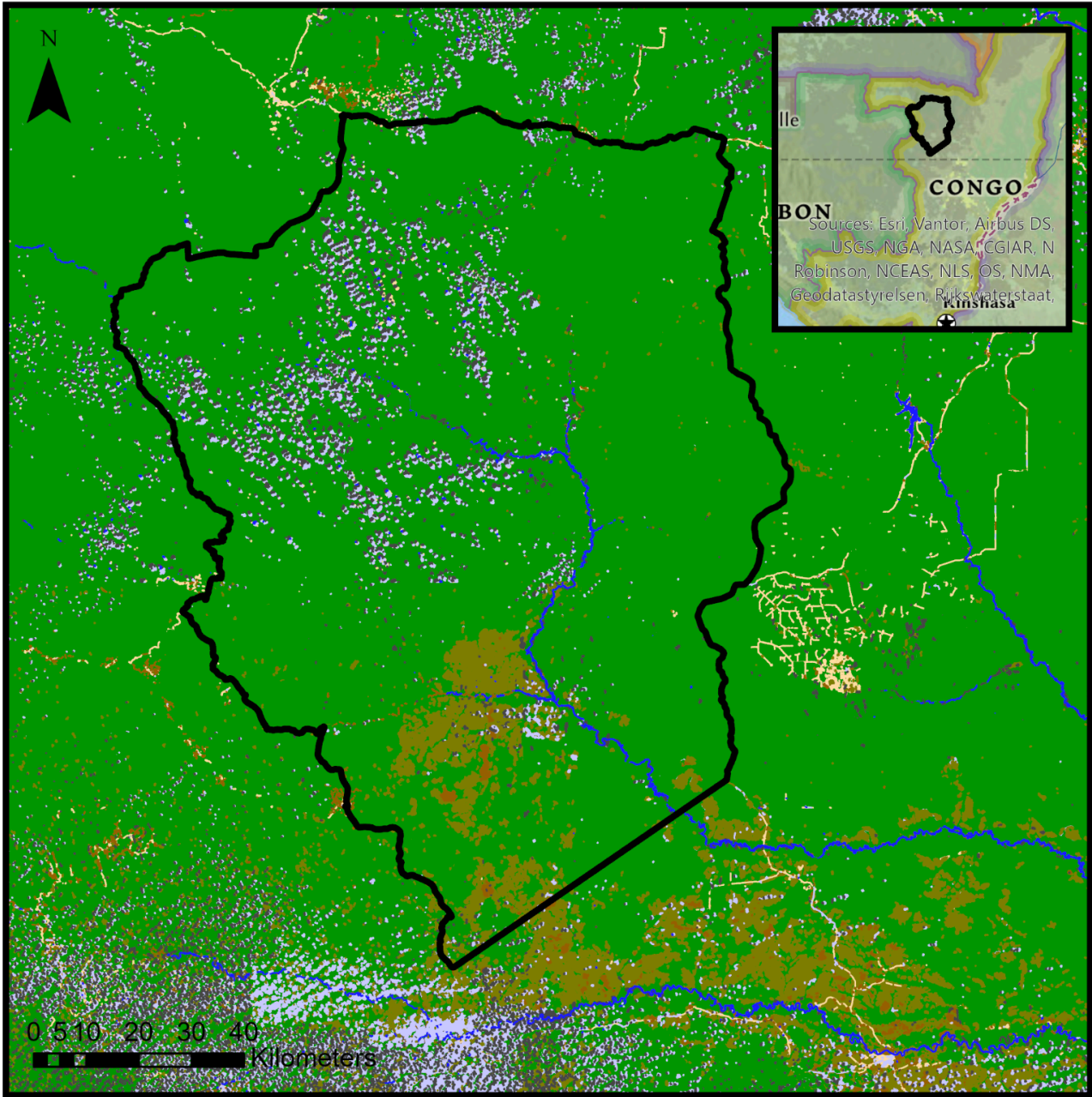
The time-series classification confirms that the primary mining operation has nearly quadrupled in size over the last nine years, driving a direct and corresponding loss of over 2.73 square kilometers of primary forest. More alarming is the scale and trajectory: a 261.6% increase in mining area in under a decade represents a serious and accelerating threat to one of the world's most important carbon sinks and biodiversity reserves. Agricultural encroachment, while a secondary concern, is widespread at the park margins and closely tied to the road network. These findings provide an important foundation for conservation planning and monitoring efforts in Odzala-Kokoua National Park.

Data Access & Deliverables:

<https://byu.box.com/s/vtzox8seyrb42wqbo8rdsikw8yatt7j1>

- **OKNP_Report_2026.pdf** - Final project report
- **Odzala_Presentation_2026.pptx** - PowerPoint presentation of findings
- **Imagery** - Sentinel-2 scenes and Planet basemaps
- **Training_Data** - polygons used for training and additional relevant polygons
- **Object_Based_Classification** - Contains our S2_Tiles_Object classification and various relevant eCognition files
- **Pixel_Based_Classification** - Our pixel-based classification results as well as any other relevant processed raster files
- **Supporting vector/raster files** - Supplementary files that may be useful in additional or other analysis

Odzala-Kokoua National Park Object-Based Land Cover Classification

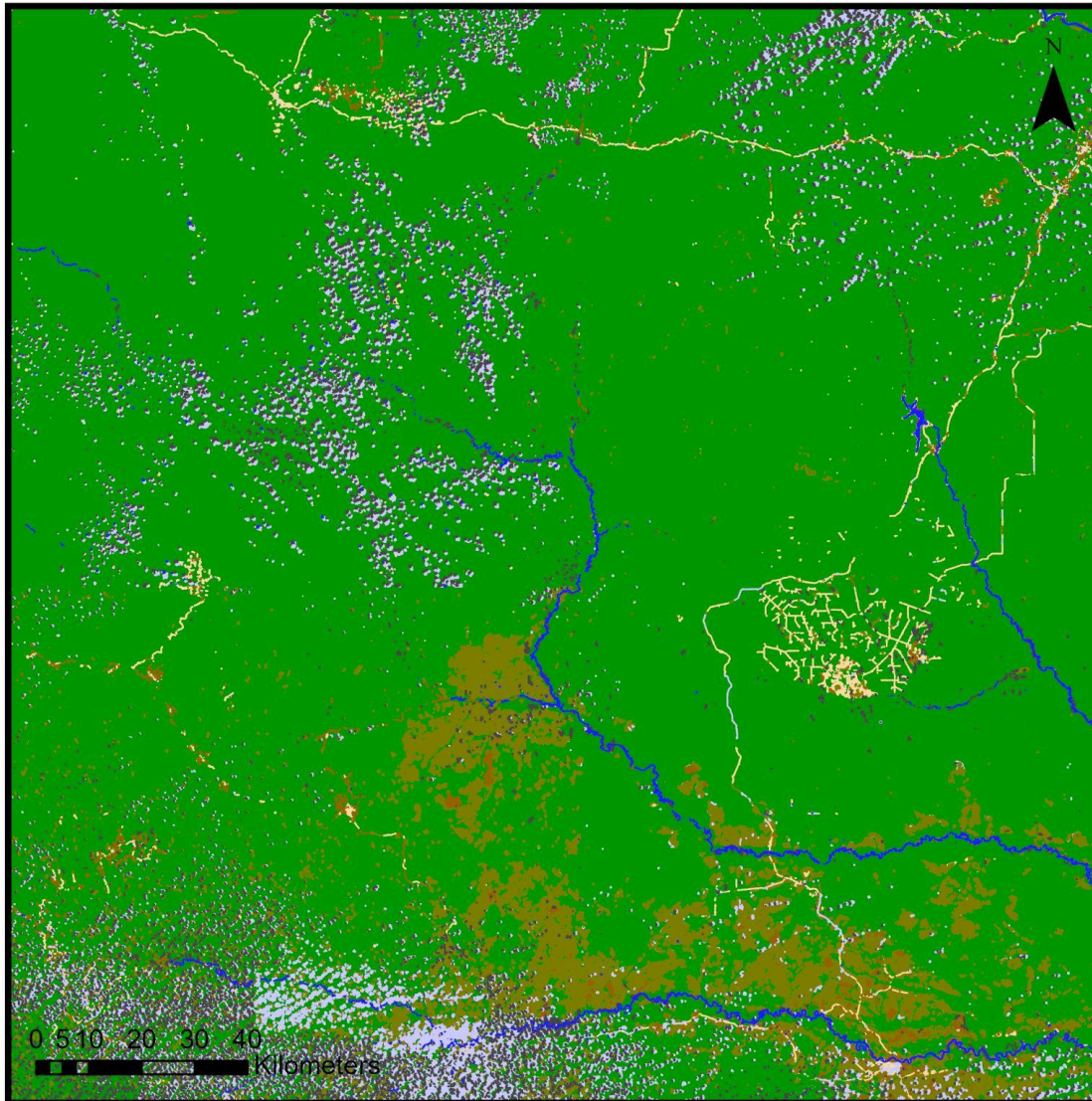


Classification map created with eCognition nearest neighbor algorithm using Sentinel-2 data from April 4, 2025 and December 12, 2025

Land Cover Classification

- | | |
|---------------|---------|
| barren_bright | savanna |
| barren_dark | shadows |
| clouds | water |
| forest | |

Odzala-Kokoua National Park Object-Based Land Cover Classification



Classification map created with eCognition nearest neighbor algorithm using Sentinel-2 data from April 4, 2025 and December 12, 2025

Land Cover Classification

barren_bright	savanna
barren_dark	shadows
clouds	water
forest	